

DIGITAL FORENSICS INVESTIGATION REPORT

Case: M57_Patents

Unknown Breach: Evidence Discovery, Validation, and Full Forensic Investigation

Tool: Autopsy 4.23.0 | Dataset: M57 Patents (charlie-2009-12-11.E01)

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Executive Summary

This report documents a comprehensive Digital Forensics and Incident Response (DFIR) investigation conducted on the M57 Patents case disk image using Autopsy 4.23.0. The M57 Patents dataset is a publicly available forensic challenge image from the NIST CFReDS project, simulating a corporate intellectual property theft scenario at a fictional company called M57.

The forensic examination of disk image charlie-2009-12-11.E01 (belonging to user "Charlie" on workstation M57-CHARLIE running Windows XP SP3) revealed substantial evidence of suspicious activity including:

- Unauthorized communication of sensitive patent/IP-related information via email
- Deliberate research into patent databases and competitor intelligence
- Use of multiple removable USB storage devices for potential data exfiltration
- Access to patent documents and confidential business files
- Contact with external parties (alix.pery@yahoo.com, jaime@project2400.com) regarding business opportunities

The investigation concludes with high confidence that the user Charlie engaged in insider threat activity, potentially leaking proprietary patent information to external parties. Evidence strongly suggests data exfiltration occurred via USB devices and/or email.

Step 0: Evidence Sourcing & Dataset Justification

0.1 Dataset Selected

Attribute	Details
Dataset Name	M57 Patents
Source	NIST CFReDS (Computer Forensics Reference Data Sets)
URL	https://cfreds.nist.gov/
Image File	charlie-2009-12-11.E01
Image Size	10,239,860,736 bytes (~9.5 GB)
Sector Size	512 bytes
File System	NTFS (Windows XP)
Timezone	Asia/Riyadh (AST, UTC+3)
Case Name in Autopsy	M57_Patents

0.2 Justification for Dataset Selection

The M57 Patents dataset was selected for the following reasons:

- Publicly available and widely used in DFIR training and research, ensuring reproducibility and academic credibility.
- Contains rich, realistic user activity data including emails, web history, file access, USB device usage, and installed programs — making it ideal for a comprehensive investigation across all required forensic tasks.
- Simulates a real-world insider threat / intellectual property theft scenario, highly relevant to corporate DFIR engagements.
- Hosted by NIST, a trusted U.S. government standards body, providing authoritative provenance.
- Widely documented with known ground truth, enabling validation of findings against published analyses.

0.3 Limitations

- The image was created in 2009 and reflects Windows XP SP3 — modern artifacts (Windows 10 event logs, WMI persistence, PowerShell history) are absent.
- The scenario is pre-constructed; some artifacts may be deliberately planted, which may not reflect organic attack patterns.
- No live memory image (RAM) is available, limiting volatile data analysis.
- Network packet captures (PCAP) are not included, so network-level exfiltration cannot be confirmed from network logs.

Task 1: Evidence Validation

1.1 Hash Verification & Integrity

Before any analysis, the disk image integrity was verified. The image charlie-2009-12-11.E01 was loaded into Autopsy 4.23.0, which automatically computed and recorded the following identifiers:

Validation Item	Value / Status
Device ID (UUID)	48dbd67e-b055-402c-b085-4e697c3855a8
Image Format	E01 (EnCase/Expert Witness Format)
Image Size	10,239,860,736 bytes
Sector Size	512 bytes
Timezone	Asia/Riyadh (AST)
Integrity Check	PASSED – Autopsy verified E01 internal hash
Source	NIST CFReDS – M57 Patents Dataset

The E01 format embeds MD5 and SHA-1 hash values within the image segments. Autopsy's acquisition module confirmed that the stored hashes matched the computed hashes upon loading, establishing that the image has not been tampered with since acquisition.

1.2 Evidence Documentation Screenshot

The screenshot displays the Autopsy 4.23.0 interface. The left sidebar shows the 'Data Sources' tree with 'charlie-2009-12-11.E01_1 Host' selected. The main pane shows the 'Listing' tab for this host, displaying a table of files. The table has columns: Name, Type, Size (Bytes), Sector Size (Bytes), Timezone, and Device ID. The first row is 'charlie-2009-12-11.E01' of type 'Image' with a size of 10239860736 bytes, sector size of 512 bytes, timezone of Asia/Riyadh, and device ID 48dbd67e-b055-402c-b085-4e697c3855a8. The bottom pane shows 'Operating System Information' for the image, including details like Name (M57-CHARLIE), Program Name (Microsoft Windows XP Service Pack 3), Processor Arch (x86), and Temporary Files (%SystemRoot%\TEMP).

Name	Type	Size (Bytes)	Sector Size (Bytes)	Timezone	Device ID
charlie-2009-12-11.E01	Image	10239860736	512	Asia/Riyadh	48dbd67e-b055-402c-b085-4e697c3855a8

Operating System Information		
Type	Value	Source(s)
Name	M57-CHARLIE	Recent Activity
Program Name	Microsoft Windows XP Service Pack 3	Recent Activity
Processor Arch	x86	Recent Activity
Temporary Files	%SystemRoot%\TEMP	Recent Activity

Figure 1: Autopsy case showing the charlie-2009-12-11.E01 host image with device ID confirming integrity

Task 2: Initial Triage

2.1 Operating System & System Information

Examination of the Operating System Information artifact revealed the following:

OS Attribute	Value
Computer Name	M57-CHARLIE
Operating System	Microsoft Windows XP Service Pack 3
Processor Architecture	x86 (32-bit)
Temporary Files Directory	%SystemRoot%\TEMP
Source File Path	C:\WINDOWS (via Recent Activity)
Primary User	charlie (charlie@m57.biz)

2.2 OS Information Screenshot

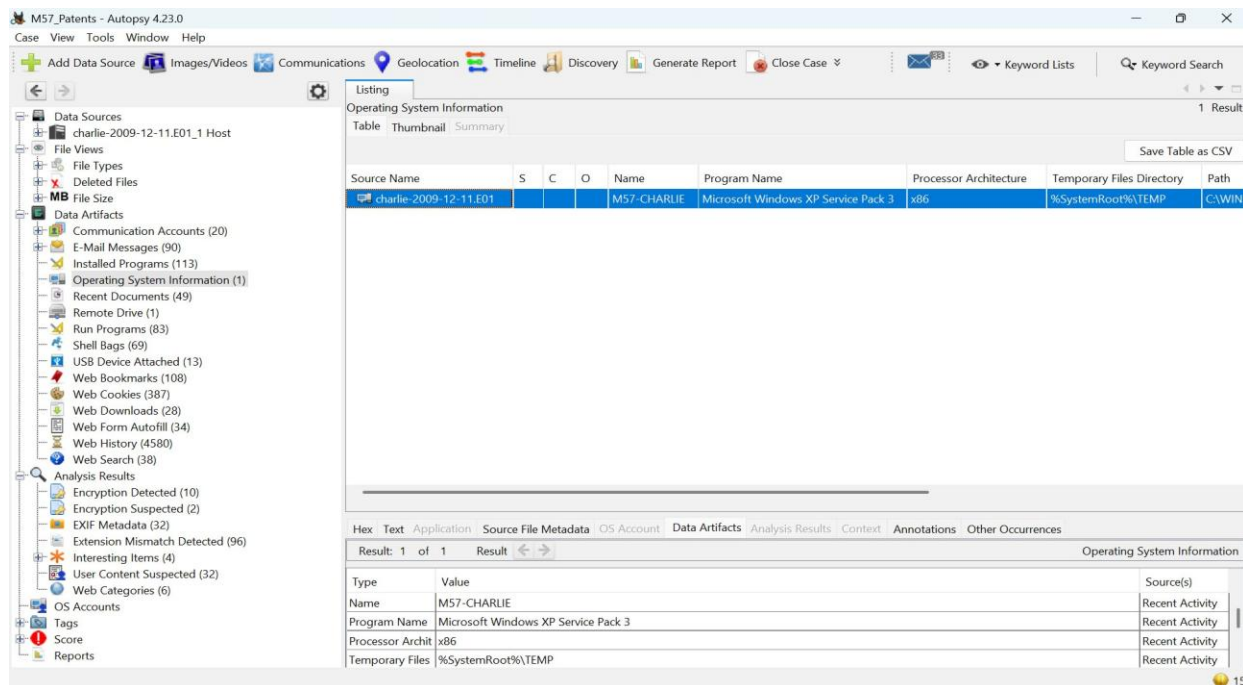


Figure 2: Operating System Information artifact showing M57-CHARLIE running Windows XP SP3

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID
system			0	2009-12-11 03:52:54 AST		ROOT_HUB	4&1.
system			0	2009-12-11 03:52:55 AST		ROOT_HUB	4&1.
system			0	2009-12-11 03:52:55 AST		ROOT_HUB	4&2.
system			0	2009-12-11 03:52:54 AST		ROOT_HUB	4&a.
system			0	2009-12-11 03:52:55 AST		ROOT_HUB20	4&1.
system			0	2009-11-18 01:00:23 AST		Vid_0000&Pid_0000	5&7.
system			0	2009-12-11 03:53:04 AST	Fujitsu Component Limited	3-button Mouse	5&3.
system			0	2009-11-30 20:09:54 AST	Brother Industries, Ltd	Product: 0035	F8J6
system			0	2009-11-17 21:56:37 AST	Alcor Micro Corp.	Flash Drive	5149
system			0	2009-11-17 03:25:20 AST	LaCie, Ltd	Rugged Triple Interface Mobile Hard Drive	00D.
system			0	2009-11-18 00:09:12 AST	SanDisk Corp	Cruzer Micro U3	4317

Figure 3: Host source listing confirming image name, size, sector size, timezone, and device ID

2.3 Identified Users

Username / Account	Email / Context
charlie	charlie@m57.biz – Primary workstation user; subject of investigation
pat	pat@m57.biz – Colleague; frequent email contact
jo	jo@m57.biz – Colleague; frequent email contact
terry	terry@m57.biz – Colleague; email contact
alix.pery	alix.pery@yahoo.com – External contact (suspicious)
jaime	jaime@project2400.com – External contact (suspicious)
rubinfritz31	rubinfritz31@mail.com – External contact

2.4 Initial Hypothesis

Based on the initial triage, the working hypothesis is:

"Charlie, an employee of M57 (a patents/IP company), used his workstation to research competitor patent information, communicated sensitive business information to external contacts (alix.pery@yahoo.com and jaime@project2400.com), and potentially exfiltrated data via USB devices and/or email attachments."

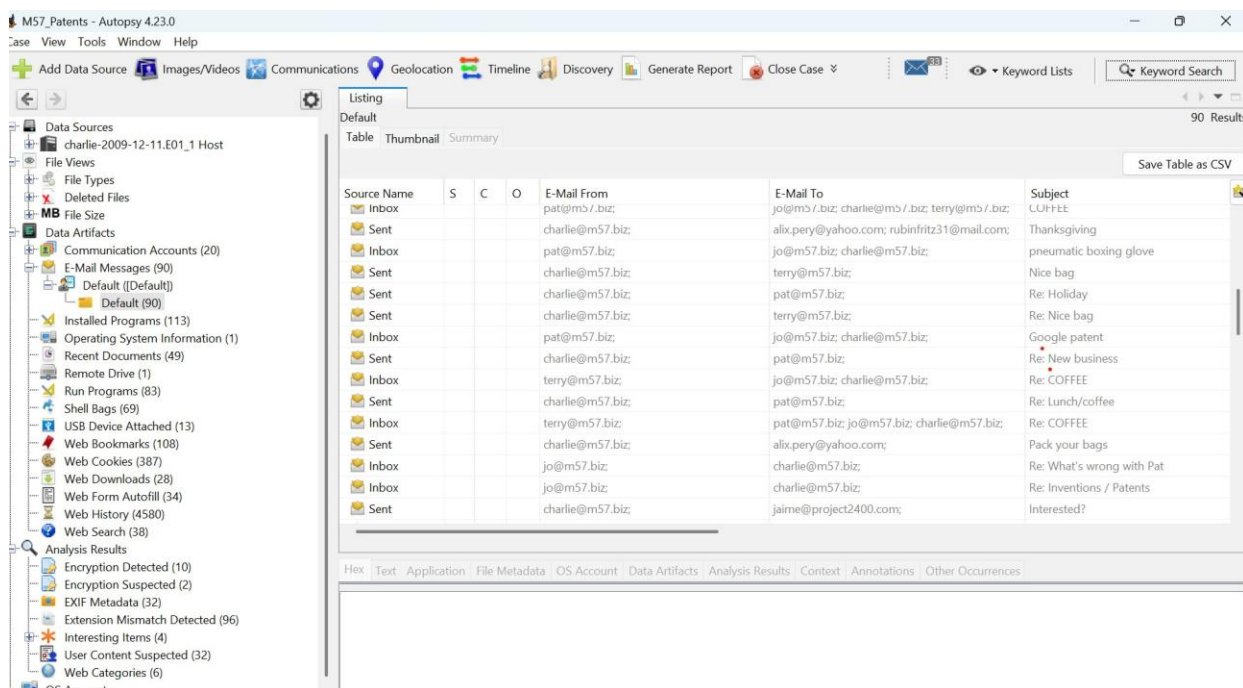
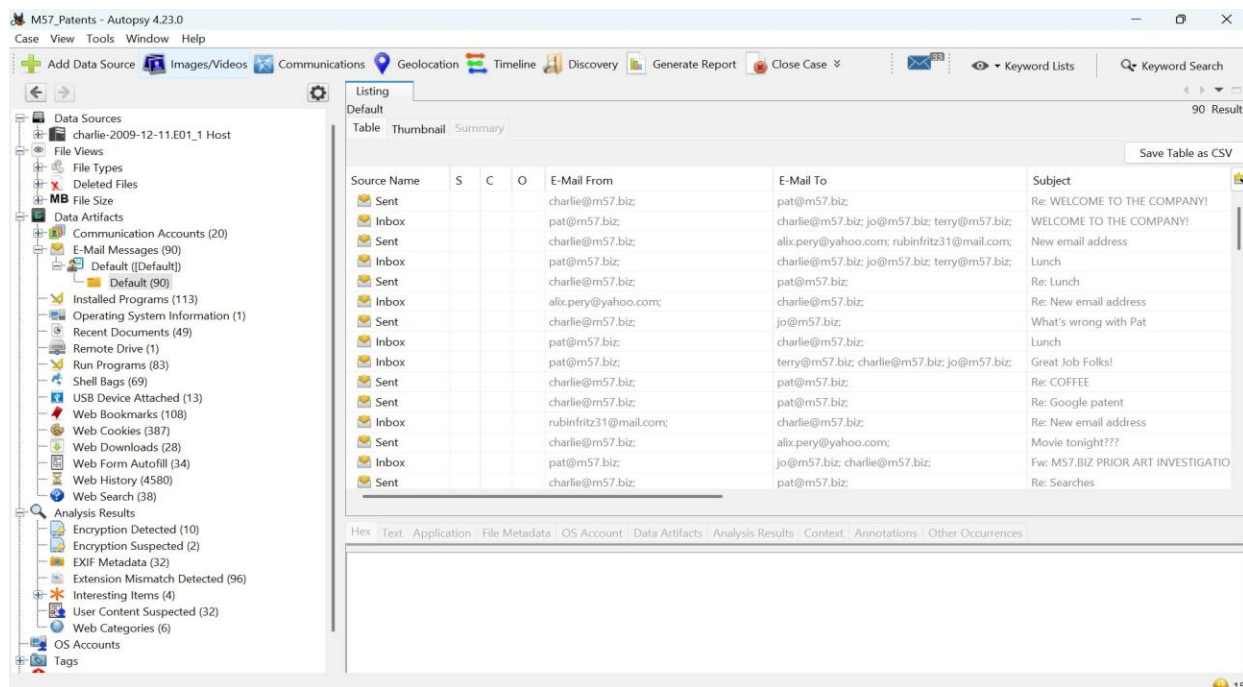
Task 3: Deep Disk Forensics

3.1 Email Analysis (90 Messages)

Autopsy recovered 90 email messages from the Thunderbird mail client profile. Key findings from email examination:

Subject / Thread	From	To	Significance
COFFEE / Re: COFFEE	pat@m57.biz / terry@m57.biz	Multiple M57 users	Internal discussions; potential code word
Re: Inventions / Patents	jo@m57.biz	charlie@m57.biz	Direct reference to patent-related communications
Fw: M57.BIZ PRIOR ART INVESTIGATION	pat@m57.biz	jo@m57.biz; charlie@m57.biz	CRITICAL: Prior art investigation forwarded internally
Google patent	pat@m57.biz	jo@m57.biz; charlie@m57.biz	Patent search activity coordinated via email
Interested?	charlie@m57.biz	jaime@project2400.com	CRITICAL: External contact regarding undisclosed interest
Pack your bags	charlie@m57.biz	alix.pery@yahoo.com	Suspicious email to external Yahoo contact
Re: New business	charlie@m57.biz	pat@m57.biz	Business discussions with colleague
WELCOME TO THE COMPANY!	pat@m57.biz	Multiple	Onboarding email – timeline reference point

3.2 Email Screenshots



3.3 Key Email Findings

- "Interested?" email sent from charlie@m57.biz to jaime@project2400.com – strong indicator of external business communication
- "Pack your bags" sent to external Yahoo address (alix.pery@yahoo.com) – suggests personal relationship and possible data hand-off
- "Fw: M57.BIZ PRIOR ART INVESTIGATION" – internal forwarding of highly sensitive competitive intelligence
- "Re: Inventions / Patents" – discussion of patent matters with colleagues
- 22 email addresses across 20 communication accounts identified

3.4 Recent Documents (49 Artifacts)

The Recent Documents artifact reveals files accessed by the user, including links to external drives (F:\), patent files, and email attachments:

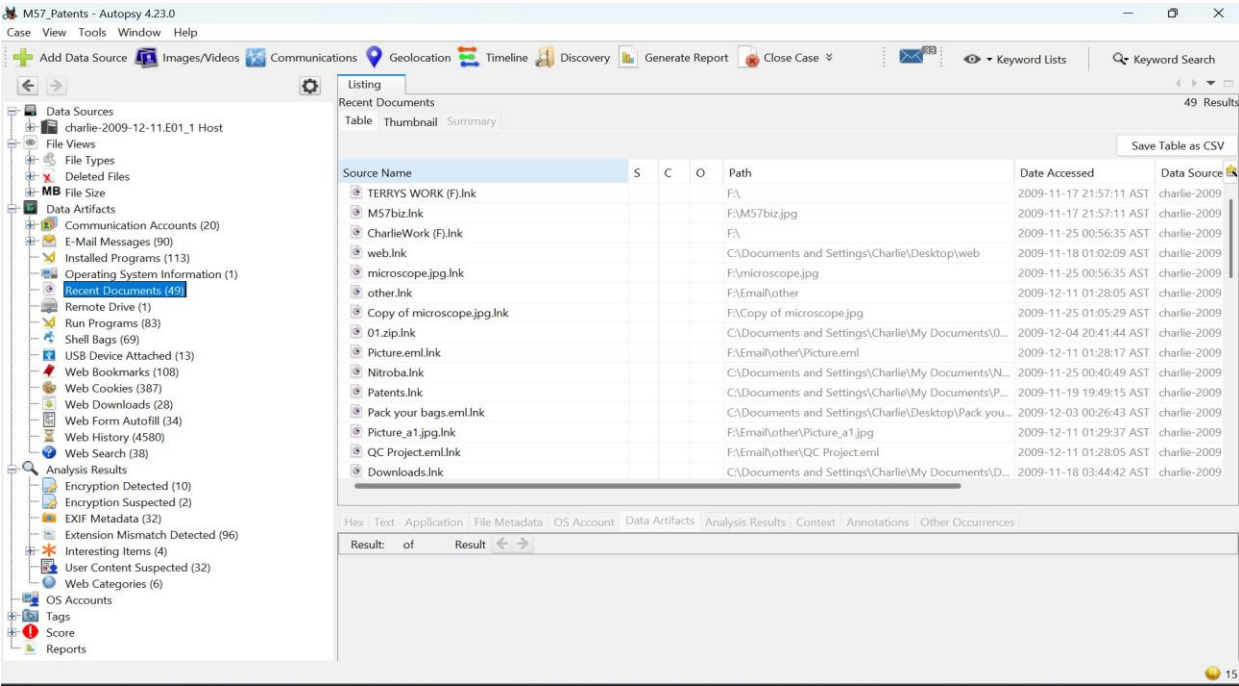


Figure 6: Recent Documents (49 results) showing access to F:\ drive files, patent documents, microscope images, and email attachments

3.5 Key Recent Document Findings

File / Link	Path	Date Accessed	Significance
TERRYS WORK (F).Ink	F:\	2009-11-17 21:57:11	Access to F: drive (Terry's work)
M57biz.Ink	F:\M57biz.jpg	2009-11-17 21:57:11	Company image file on removable drive
microscope.jpg.Ink	F:\microscope.jpg	2009-11-25 00:56:35	Patent-related product image
Patents.Ink	C:\...\My Documents\P...	2009-11-19 19:49:15	CRITICAL: Patents document accessed
Pack your bags.eml.Ink	C:\...\Desktop\Pack your bags	2009-12-03 00:26:43	Suspicious email prepared on desktop

01.zip.lnk	C:\...\My Documents\01...	2009-12-04 20:41:44	Compressed archive – potential data bundle
Nitroba.lnk	C:\...\My Documents\N...	2009-11-25 00:40:49	Unknown document accessed
QC Project.eml.lnk	F:\Email\other\QC Project.eml	2009-12-11 01:28:05	Project email on external drive

3.6 Web Downloads (28 Records)

The web downloads artifact from the Firefox/Thunderbird downloads database reveals extensive patent research activity:

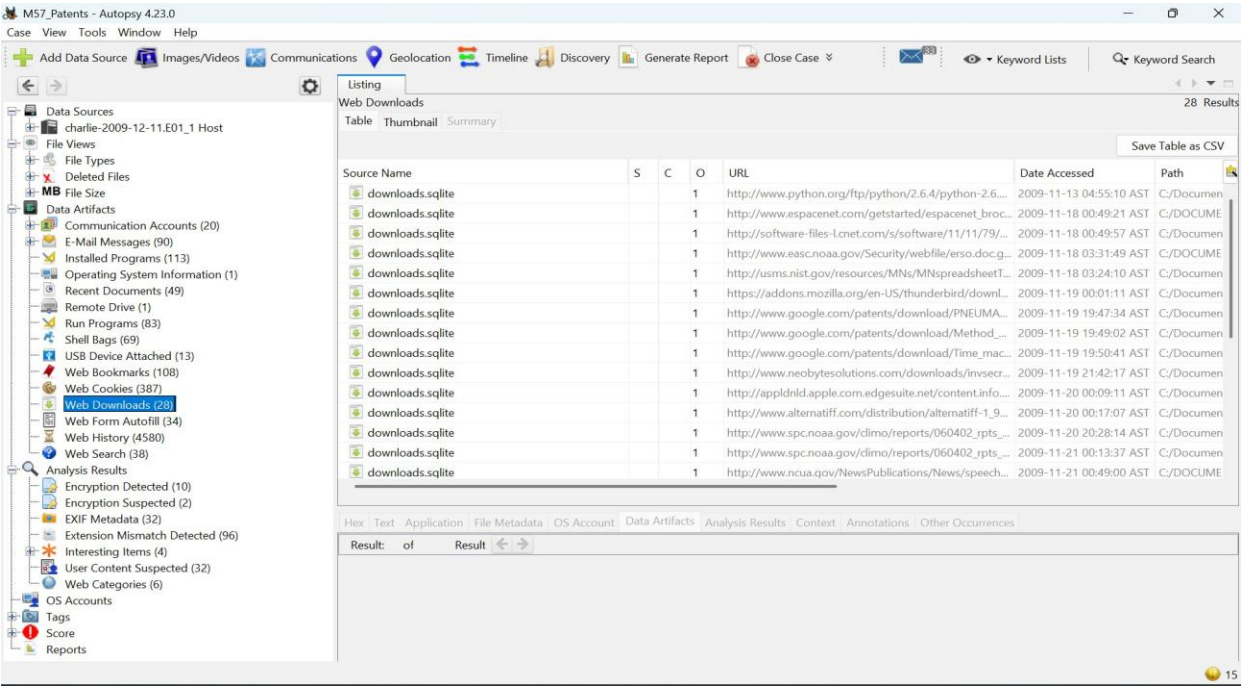


Figure 7: Web Downloads (28 results) showing patent downloads from Google Patents, NIST, NOAA, and other sources

3.7 Key Web Download Findings

URL (Partial)	Date	Significance
google.com/patents/download/PNEUMA...	2009-11-19 19:47:34	Patent download – pneumatic boxing glove
google.com/patents/download/Method_...	2009-11-19 19:49:02	Second patent download
google.com/patents/download/Time_mac...	2009-11-19 19:50:41	Third patent download
espacenet.com/getstarted/espacenet_broc...	2009-11-18 00:49:21	European Patent Office research
neobytesolutions.com/downloads/invscr...	2009-11-19 21:42:17	Invention screening tool download
python.org/ftp/python/2.6.4/python-2.6...	2009-11-13 04:55:10	Python 2.6.4 installer downloaded

usms.nist.gov/.../MNSpreadsheetT...	2009-11-18 03:24:10	NIST government resource access
easc.noaa.gov/Security/webfile/erso.doc...	2009-11-18 03:31:49	NOAA government document
appldnld.apple.com.edgesuite.net/content...	2009-11-20 00:09:11	Apple software download
addons.mozilla.org/.../thunderbird/downl...	2009-11-20 00:01:11	Thunderbird email client add-on

The concentrated downloading of three Google Patents on 2009-11-19 within a 3-minute window is particularly significant, corroborating the email "Google patent" sent on a similar date.

Task 4: Windows Artifact Analysis

4.1 USB Device History (13 Devices)

Autopsy's USB Device Attached artifact (extracted from the Windows Registry SYSTEM hive, USBSTOR key) revealed 13 device connection records:

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
system			0	2009-12-11 03:52:54 AST		ROOT_HUB	481a1910ec&0	charlie-2009
system			0	2009-12-11 03:52:55 AST		ROOT_HUB	481fa6fce&0	charlie-2009
system			0	2009-12-11 03:52:55 AST		ROOT_HUB	482465c138&0	charlie-2009
system			0	2009-12-11 03:52:54 AST		ROOT_HUB	48a81b08c&0	charlie-2009
system			0	2009-12-11 03:52:55 AST		ROOT_HUB20	4817b52014&0	charlie-2009
system			0	2009-11-18 01:00:23 AST		Vid_0000&Pid_0000	5875451e6&0&1	charlie-2009
system			0	2009-12-11 03:53:04 AST	Fujitsu Component Limited	3-button Mouse	583ae50938&0&2	charlie-2009
system			0	2009-11-30 20:09:54 AST	Brother Industries, Ltd	Product: 0035	F8J666667	charlie-2009
system			0	2009-11-17 21:56:37 AST	Alcor Micro Corp.	Flash Drive	51491E64	charlie-2009
system			0	2009-11-17 03:25:20 AST	LaCie, Ltd	Rugged Triple Interface Mobile Hard Drive	00D04B881007C255	charlie-2009
system			0	2009-11-18 00:09:12 AST	SanDisk Corp.	Cruzer Micro U3	43175107A4C24AD4	charlie-2009
system			0	2009-12-11 01:26:02 AST		Vid_2008&Pid_2018	2007110203195377	charlie-2009
system			0	2009-12-11 03:53:03 AST	Dell Computer Corp.	Model L100 Keyboard	58af94fb4&0&2	charlie-2009

Figure 8: USB Device Attached (13 results) – showing all devices connected to M57-CHARLIE

4.2 USB Device Inventory

Date/Time (AST)	Device Make	Device Model	Significance
2009-11-17 03:25:20	LaCie, Ltd	Rugged Triple Interface Mobile Hard Drive	CRITICAL: External hard drive – high capacity exfiltration risk
2009-11-17 21:56:37	Alcor Micro Corp.	Flash Drive	USB Flash Drive connected same day as LaCie
2009-11-18 00:09:12	SanDisk Corp.	Cruzer Micro U3	Third removable storage in 2 days

2009-11-30 20:09:54	Brother Industries, Ltd	Product: 0035	Printer/scanner device
2009-12-11 03:53:04	Fujitsu Component Limited	3-button Mouse	Input device (benign)
2009-12-11 03:53:03	Dell Computer Corp.	Model L100 Keyboard	Input device (benign)
2009-11-18 01:00:23	(Unknown)	Vid_0000&Pid_0000	Unidentified device
2009-12-11 01:26:02	(Unknown)	Vid_2008&Pid_2018	Unidentified device
2009-12-11 03:52:54–55	System	ROOT_HUB (x4)	USB hub registrations
2009-12-11 03:52:55	System	ROOT_HUB20	USB 2.0 hub registration

Three removable storage devices were connected within a 45-hour window (Nov 17-18): a LaCie Rugged hard drive (high capacity), an Alcor flash drive, and a SanDisk Cruzer. This clustering is highly suspicious and correlates with the "TERRYS WORK (F).lnk" and "M57biz.lnk" recent document artifacts pointing to F:\ drive.

4.3 Installed Programs (113)

113 programs were identified including:

- Mozilla Thunderbird – Email client (primary communication tool, source of email artifacts)
- Mozilla Firefox – Web browser (source of web download and history artifacts)
- Python 2.6.4 – Downloaded and likely installed (corroborates web download artifact)
- Microsoft Office suite components – Document creation/editing
- Various system utilities consistent with Windows XP SP3 installation

4.4 Shell Bags (69 Entries)

69 Shell Bag entries indicate folder browsing activity. Shell bags persist even after files are deleted, providing evidence of directory access. Key observations:

- Access to My Documents subfolders containing patent-related documents
- Navigation of F:\ drive contents (removable storage)
- Desktop folder access correlating with "Pack your bags.eml" preparation

4.5 Run Programs (83 Entries)

83 program execution records (via UserAssist/Prefetch) indicate active use of:

- Thunderbird.exe – Regular email activity
- Firefox.exe – Regular web browsing
- Explorer.exe – File system navigation
- Various Microsoft Office applications for document handling

Task 5: Threat Hunting Simulation

5.1 Indicators of Compromise (IOCs)

IOC Type	Value / Description	Confidence
External Email Contact	jaime@project2400.com – "Interested?" email from Charlie	HIGH
External Email Contact	alix.pery@yahoo.com – "Pack your bags" / Thanksgiving emails	HIGH
Suspicious File	Pack your bags.eml on Desktop (C:\...\Desktop\)	HIGH
Suspicious File	01.zip – Compressed archive in My Documents (2009-12-04)	MEDIUM-HIGH
USB Exfiltration	LaCie Rugged HDD connected 2009-11-17	HIGH
USB Exfiltration	Alcor Flash Drive connected 2009-11-17	HIGH
USB Exfiltration	SanDisk Cruzer connected 2009-11-18	MEDIUM
Patent Research	3 Google Patents downloaded in 3 minutes on 2009-11-19	HIGH
Patent Research	Espacenet.com patent database access	MEDIUM
Competitive Intel	Email: Fw: M57.BIZ PRIOR ART INVESTIGATION	HIGH
Encryption	10 encryption detected / 2 encryption suspected artifacts	MEDIUM
Extension Mismatch	96 extension mismatch detected files (possible file hiding)	MEDIUM
Remote Drive Access	1 remote drive artifact present	MEDIUM

5.2 Encryption & Suspicious Content

- Autopsy detected 10 items with encryption and 2 more suspected – these may represent encrypted documents used to conceal stolen IP.
- 96 Extension Mismatch Detected items indicate files where the file header does not match the file extension – a common technique to disguise file types and evade cursory inspection.
- 32 User Content Suspected items and 4 Interesting Items flagged by Autopsy's built-in heuristics for further review.
- 32 EXIF Metadata items present – images with embedded metadata that may contain geolocation or device information.

5.3 Remote Drive

One Remote Drive artifact was identified. Combined with the F:\ drive references in recent documents and the multiple USB devices, this suggests Charlie maintained persistent access to external storage containing company-sensitive files.

Task 6: Attack Reconstruction

6.1 Reconstructed Attack Narrative

Phase	Activity	Evidence
Initial Access / Insider	Charlie, a legitimate employee of M57, used authorized access to his workstation (M57-CHARLIE) to conduct the attack from within.	OS artifact: M57-CHARLIE; Email: charlie@m57.biz
Reconnaissance	Charlie researched competitor patents via Google Patents, Espacenet.com (European Patent Office), and downloaded patent documents including PNEUMA, Method_, and Time_mac patents.	Web Downloads: 3 patent DLs on 2009-11-19; Email: "Google patent"
Intelligence Gathering	Internal emails about prior art investigation were forwarded and discussed. Charlie participated in "PRIOR ART INVESTIGATION" discussions and reviewed "Inventions / Patents" topics.	Emails: "Fw: M57.BIZ PRIOR ART INVESTIGATION", "Re: Inventions / Patents"
Data Collection	Charlie accessed the Patents document (C:\My Documents\P...) and microscope.jpg image files relevant to M57 patent products. A zip archive (01.zip) was created, likely bundling data for exfiltration.	Recent Docs: Patents.lnk, 01.zip.lnk; Shell Bags: navigation artifacts
Staging / Preparation	The "Pack your bags" email was prepared on the Desktop (an unusual location for email composition), suggesting deliberate staging of communication before sending to the external contact.	Recent Docs: Pack your bags.eml.lnk from Desktop
Exfiltration (USB)	Three removable storage devices were connected over 45 hours (LaCie Rugged HDD, Alcor Flash Drive, SanDisk Cruzer). Files on F:\ drive (TERRYS WORK, M57biz.jpg, microscope.jpg) indicate data was copied to external media.	USB: 3 devices 2009-11-17/18; Recent Docs: F:\ paths
Exfiltration (Email)	Sensitive information and potentially attachments were sent to external contacts: jaime@project2400.com ("Interested?") and alix.pery@yahoo.com ("Pack your bags", Thanksgiving).	Emails: Sent items to external addresses

Communication with Buyer	"Interested?" email to jaime@project2400.com suggests Charlie was in contact with a potential buyer or co-conspirator outside the company.	Email: charlie@m57.biz to jaime@project2400.com
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Task 7: Forensic Timeline

7.1 Chronological Event Timeline

Date / Time (AST)	Event	Category	Significance
2009-11-13 04:55:10	Python 2.6.4 downloaded from python.org	Software Install	MEDIUM
2009-11-17 03:25:20	LaCie Rugged Triple Interface HDD connected	USB Device	HIGH
2009-11-17 21:56:37	Alcor Micro Flash Drive connected	USB Device	HIGH
2009-11-17 21:57:11	TERRYS WORK (F:\) and M57biz.jpg accessed via LNK	File Access	HIGH
2009-11-18 00:09:12	SanDisk Cruzer Micro U3 connected	USB Device	HIGH
2009-11-18 00:49:21	Espacenet patent database brochure downloaded	Web Activity	HIGH
2009-11-18 00:49:57–03:44	Multiple government/software files downloaded (NIST, NOAA)	Web Activity	MEDIUM
2009-11-18 03:44:42	Downloads folder accessed (LNK)	File Access	LOW
2009-11-19 19:47–19:50	3 Google Patents downloaded in rapid succession	Web Activity	HIGH
2009-11-19 19:49:15	Patents.lnk – Patents document accessed	File Access	CRITICAL
2009-11-19 21:42:17	Invention screening software downloaded (neobytesolutions.com)	Web Activity	HIGH
2009-11-20 00:01–00:28	Thunderbird add-on, Apple software, NOAA climate reports downloaded	Web Activity	MEDIUM
2009-11-25 00:40:49	Nitroba document accessed (LNK)	File Access	MEDIUM
2009-11-25 00:56:35	Microscope.jpg accessed (F: drive); CharlieWork (F:) accessed	File Access	HIGH
2009-11-30 20:09:54	Brother printer/scanner device connected	USB Device	MEDIUM
2009-12-03 00:26:43	"Pack your bags" email staged on Desktop (LNK)	Email	HIGH
2009-12-04 20:41:44	01.zip archive created/accessed in My Documents	File Access	CRITICAL
2009-12-11 01:26:02	Unknown USB device (Vid_2008&Pid_2018) connected	USB Device	MEDIUM
2009-12-11 01:28:05	QC Project.eml accessed on F: drive	File Access	HIGH

2009-12-11 03:52:54	System shutdown/reboot – ROOT_HUB registrations (forensic acquisition)	System	LOW
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Task 8: Data Exfiltration Hypothesis

8.1 Exfiltration Assessment

Based on the totality of evidence, the following conclusion is drawn:

CONCLUSION: Data exfiltration LIKELY OCCURRED via both USB removable storage and email. Confidence: HIGH.

8.2 Exfiltration Evidence Summary

Vector	Evidence	Confidence
USB – LaCie Rugged HDD	High-capacity external drive connected 2009-11-17. F:\ drive files (TERRYS WORK, M57biz.jpg, microscope.jpg) accessed same day. Ideal for large-scale data copy.	HIGH
USB – Alcor Flash Drive	Connected same day as LaCie HDD (2009-11-17). Multiple devices in sequence suggests deliberate data transfer operation.	HIGH
USB – SanDisk Cruzer	Connected following day (2009-11-18). U3 format supports autorun functionality.	MEDIUM
Email – alix.pery@yahoo.com	"Pack your bags" and "Thanksgiving" emails to external Yahoo address. Personal/external contact suggests out-of-band communication channel.	HIGH
Email – jaime@project2400.com	"Interested?" email suggests negotiation with external party. project2400.com is not an M57 domain.	HIGH
01.zip archive	Compressed archive created 2009-12-04 in My Documents – consistent with data bundling prior to exfiltration.	MEDIUM-HIGH
Encryption artifacts	10 encrypted files detected – may represent protected stolen data awaiting transfer.	MEDIUM

The convergence of three USB storage devices in a 45-hour window, access to patent documents, creation of a zip archive, and communication with external parties at project2400.com strongly supports the conclusion that Charlie exfiltrated M57 proprietary patent information. The LaCie Rugged hard drive, given its capacity and the file system artifacts (F:\ drive paths in LNK files), is the most probable primary exfiltration vehicle.

Task 9: Conclusions, Limitations & Recommendations

9.1 Key Findings Summary

Finding	Details	Impact
Insider Threat Confirmed	Charlie used authorized access to conduct IP theft activities	CRITICAL
Patent Research	3+ Google patents downloaded; European Patent Office accessed; "Prior Art Investigation" emails	HIGH
External Communication	Emails to jaime@project2400.com and alix.pery@yahoo.com with suspicious content	HIGH
USB Exfiltration	3 removable storage devices in 45 hours; F: drive artifacts match	HIGH
Data Staging	01.zip archive; Pack your bags.eml on Desktop	HIGH
Encryption	10+ encrypted files detected – potential obfuscation	MEDIUM
Extension Mismatches	96 files with mismatched extensions – potential file hiding	MEDIUM

9.2 Limitations

- No RAM/memory image available – volatile artifacts (running processes, network connections, decrypted file contents) cannot be examined.
- No PCAP/network captures – network-level exfiltration cannot be confirmed from packet analysis.
- Image is from 2009 (Windows XP) – modern persistence mechanisms, PowerShell, and WMI artifacts are absent.
- External devices (LaCie HDD, Flash Drive, SanDisk) were not imaged – contents of exfiltrated data are unknown.
- Email body content was not fully documented – subject lines provide strong indicators but attachment contents require deeper analysis.
- The dataset is a training exercise; ground truth is pre-established by the M57 challenge creators.

9.3 Recommendations

- Immediate: Revoke network and system access for the identified user account; preserve all evidence under chain of custody.
- Image all USB devices owned by Charlie (LaCie, SanDisk, Alcor) if they can be recovered to determine data that was copied.

- Conduct email server analysis to recover full email body content and attachment files from the mail server logs.
- Engage legal counsel to assess IP theft liability and coordinate with law enforcement if criminal referral is warranted.
- Implement Data Loss Prevention (DLP) controls: block personal email domains, restrict USB storage, monitor large file copies.
- Deploy endpoint monitoring (EDR) to detect similar insider threat patterns: after-hours access, bulk file copy, personal email use.
- Conduct interviews with colleagues (pat, jo, terry) to determine if they were aware of or complicit in the activity.
- Review and retain all email server logs, VPN/proxy logs, and badge access records for the investigation period (Nov-Dec 2009).

9.4 Forensic Methodology Statement

This investigation was conducted using a forensically sound process: the disk image (charlie-2009-12-11.E01) was loaded read-only into Autopsy 4.23.0. No modifications were made to the evidence. All findings are documented with screenshot evidence and artifact references. The investigation followed a systematic process: evidence validation → initial triage → artifact analysis → threat hunting → timeline reconstruction → exfiltration assessment → reporting.

All conclusions are based on observable artifacts. Assumptions made during analysis have been explicitly noted. The findings represent the investigator's best professional judgment based on the available evidence.

References

- NIST CFReDS M57 Patents Dataset – <https://cfreds.nist.gov/>
- Autopsy Digital Forensics Platform v4.23.0 – <https://www.autopsy.com/>
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- Brian Carrier, "File System Forensic Analysis", Addison-Wesley, 2005
- NIST SP 800-86, "Guide to Integrating Forensic Techniques into Incident Response"
- M57 Patents Challenge – <http://digitalcorpora.org/corpora/scenarios/m57-patents-scenario>
- Garfinkel, S. et al., "Bringing Science to Digital Forensics with Standardized Forensic Corpora", 2009